

Virginia Mwega

FULL-STACK & AI ENGINEER

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PROFILE

Full-Stack and AI Engineer who ships production AI systems end-to-end — multi-agent pipelines, retrieval-augmented generation, Claude tool use, structured outputs, and the safety floors and graceful fallbacks that keep them working when a model or API key fails. Four production AI products delivered end-to-end (schema, API, AI orchestration, auth, payments, CI), serving real users with measurable outcomes. Focus areas: family wellness, hiring, and ecommerce.

EXPERIENCE

Founder & Full-Stack AI Engineer — FamNest

2026 – Present

famnest-iota.vercel.app · AI wellness coach for exhausted parents

- Turned a single LLM call into an orchestrated multi-agent pipeline: a coach agent drafts a history-aware plan, a safety-reviewer agent screens each draft (ok / revise / crisis) through a bounded revise loop, and a memory layer summarizes each parent's mood, sleep, and stress trends back into the next plan.
- Built a deterministic crisis floor — a keyword guard escalates self-harm language to curated support resources, independent of the probabilistic model. End-to-end testing caught two safety bugs a code review missed (offline path had no crisis handling; the LLM classifier was non-deterministic on crisis).
- Implemented lightweight lexical RAG over a curated, sourced wellness corpus (chosen over vector embeddings for a free, serverless-appropriate footprint), with source provenance shown on every plan.
- Engineered a provider-swappable LLM layer running at zero API cost on Groq's free tier (Llama 3.3 70B) — one env var swaps to Claude or GPT — with a deterministic fallback plan when the model is unavailable.
- Hardened with Supabase row-level security on every table, a signup trigger provisioning each user, and a Postgres-function rate limiter; provenance and safety metadata stored in JSONB with no schema migration.

Stack: Next.js 16, React 19, Supabase (Auth, Postgres, RLS), Tailwind v4, Groq (Llama 3.3 70B), Vercel.

Solo Full-Stack & AI Engineer — Fit Parent Plan

2026 – Present

fit-parent-plan-platform.vercel.app · AI-powered fitness coaching for busy parents

- Owned the full product from rough idea to revenue: schema, AI agents, Stripe, custom JWT auth, CRO landing, event pipeline, and deploy.
- Shipped two production Claude agents (conversational 20-Minute Window workout planner and Pantry-to-Plate dinner agent) with streaming responses and mock fallback for zero-balance API resilience.
- Designed an event pipeline emitting structured logSession events consumed by n8n automation that drafts and sends a weekly Parent Pulse digest.
- Engineered no-auth resilience: IP-based rate limiting with x-forwarded-for parsing, a pagehide listener that auto-saves plans before browser close, and plan-pause via localStorage preserving 7-day breaks across sessions.
- Integrated Stripe Checkout with idempotent webhook handlers and a 3-question lead-qualification quiz filtering prospects before they reach the coach.
- **Outcomes:** 74 parents coached, 4.9 / 5 across 38 reviews, \$199/mo price point.

Stack: Next.js 15, TypeScript, Tailwind v4, Claude API, Stripe, JWT auth, SQLite, n8n, Playwright, Vercel.

Solo Builder — Hirely

2024 – 2025

hirely-a0lx.onrender.com · Full-stack AI hiring platform for parents and caregivers

- Designed and shipped 10+ production AI features end-to-end: conversational search grounded in the live database via Claude tool use, CV parsing into a structured ParentProfile, AI-ranked top matches, employer auto-screening, and AI-drafted empathy emails on status change.
- Made job flexibility a first-class database enum (Fixed / Flexible / Async), powering reliable filtering, matching, and AI scoring across the platform.
- Engineered AI response caching by (user, profile.parsed_at) so re-uploaded resumes invalidate cleanly without time-based staleness.
- Built graceful fallbacks for every AI surface, keeping the platform functional when credits run out; shipped weekly digest and stale-job nudge crons, CI-gated deploys via GitHub Actions, and Neon Postgres on a zero-ops stack.

Stack: Django, PostgreSQL (Neon), Claude API, Render, GitHub Actions, Resend, WhiteNoise.

Solo Full-Stack Engineer — PureNest Family

2023

purenest-family.vercel.app · Family wellness ecommerce, built from scratch (no starter template)

- Built a complete ecommerce platform end-to-end: schema, API, AI assistant, checkout, and admin tooling.
- Engineered idempotent Stripe webhook handling — processed event IDs persisted so retries never create duplicate orders, regardless of network failures.
- Integrated a Claude-powered conversational shopping assistant for contextual recommendations across 5 wellness categories; built full auth from scratch (email/password, OAuth, email verification, signed one-time reset tokens).
- Shipped a server-validated coupon engine and verified-purchase reviews with database-level constraints; hardened with locked-down security headers (CSP, HSTS, X-Frame-Options DENY) and an admin dashboard with full audit log.

Stack: Next.js 16, React 19, TypeScript, PostgreSQL, Prisma, NextAuth, Stripe, Claude API, Resend, Tailwind v4, Vitest, Vercel.

SKILLS

Languages: TypeScript, Python, SQL, HTML, CSS

Frameworks: Next.js 16, React 19, Django, Tailwind CSS v4

AI Engineering: Claude API (tool use, streaming, structured outputs); multi-agent orchestration (coach / safety-reviewer / memory); retrieval-augmented generation (lexical RAG with source provenance); provider-swappable LLM layer (Claude and open models via Groq / Llama 3.3 70B); deterministic safety floors; response caching; graceful fallback patterns; conversational agents

Backend: PostgreSQL, Supabase (Auth, Postgres, RLS), Prisma, SQLite, NextAuth, custom JWT auth, REST API design, schema-first development

Payments & Automation: Stripe Checkout, idempotent webhook handling, processed-event-ID dedupe, n8n workflows, webhook pipelines, cron schedules

DevOps: Vercel, Render, GitHub Actions, CI-gated deploys, Playwright, Jest, Vitest

Production Hardening: rate limiting, row-level security, security headers (CSP, HSTS, X-Frame-Options), email/OAuth/JWT auth flows, audit logs

EDUCATION & CERTIFICATIONS

Self-Taught Software Engineering · 2023 – Present

Independent study via shipping four production AI products end-to-end.

Full-Stack Engineering Certification · freeCodeCamp Online · 2024 – 2025

PUBLICATIONS

- Idempotent Stripe webhooks: the 30-line pattern that survives every retry — *Hashnode*, 2026.
- I asked 74 parents when they actually train — *Medium*, 2026.